

Nested Factors and Cluster Randomized Trials

PUBH 526

Design and Analysis of Randomized Trials in Public Health

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Definitions

- Factors A and B are **crossed** if every level of A occurs with every level of B

	Factor A			
Factor B	1	2	3	4
1	xx	xx	xx	xx
2	xx	xx	xx	xx
3	xx	xx	xx	xx

<i>A</i>	1			2			3			4		
	1	2	3	1	2	3	1	2	3	1	2	3
	x	x	x	x	x	x	x	x	x	x	x	x
	x	x	x	x	x	x	x	x	x	x	x	x

For example: you have a four-arm study (Factor A = treatment), and each arm has people from each income tertile (Factor B)

In this example, there are 2 participants for each combination of A and B

Definitions

- Factors A and B considered **nested** if
 - Certain levels of B **only occur** with a certain level of A
 - Allows us to arbitrarily number levels of B

<i>A</i>	1			2			3			4		
<i>B</i>	1	2	3	4	5	6	7	8	9	10	11	12
	x	x	x	x	x	x	x	x	x	x	x	x
	x	x	x	x	x	x	x	x	x	x	x	x

- Think clusters or hierarchical structures:
 - Kids (B) in classrooms (A)
 - Towns (B) in states (A)
 - Households (B) in villages (A)

Replication as a Nested Factor

- Consider a single factor CRD : $y_{ij} = \mu + \tau_i + \epsilon_{ij}$
- But now suppose we consider adding the factor Rep
- The table for design when $a = 3$ and $n = 4$ can be written

**Replication:
People within
treatments!**

Trt	1				2				3			
Rep	1	2	3	4	5	6	7	8	9	10	11	12
	x	x	x	x	x	x	x	x	x	x	x	x

- How we number replicates (an levels) is unimportant
- Brackets used to denote which factor it's nested within

$$y_{ij} = \mu + \tau_i + r_{j(i)}$$

- Replication variability is used as error, $e_{ij} \leftrightarrow r_{j(i)}$

Subsampling

EU = experimental unit

- In many problems, difficult to measure EU response
- **Subsampling** - measuring EU numerous times
- Usually combine measurements for better precision
- What if we include subsample as factor in analysis?

EU Sub	1				2				3			
	1	2	3	4	5	6	7	8	9	10	11	12
	x	x	x	x	x	x	x	x	x	x	x	x

- No association between subsamples across EUs
- Thus numbering of subsamples arbitrary
- In other words, subsamples are always a nested factor

Simple Random Effects Model

$$y_{ij} = \mu + \tau_i + \epsilon_{ij} \quad \left\{ \begin{array}{l} i = 1, 2 \\ j = 1, 2, \dots, n \end{array} \right.$$

- Primarily interested in μ or σ_τ^2
- Two stages of sampling
 - Randomly choose units of interest (σ_τ^2)
 - Obtain measurements on that unit of interest (σ^2)
- In some situations, can view measurements as subsampling
- Use subsampling variability to test $H_0 : \sigma_\tau^2 = 0$

Reasoning for Nesting

- Consider the following two examples
 - 1 Drug company interested in stability of product
 - Two manufacturing sites
 - Three batches from each site
 - Ten tablets from each batch
 - 2 Stratified random sampling procedure
 - Randomly sample five states
 - Randomly select three counties
 - Randomly select two towns
 - Randomly select five households
- More manageable experiments than factorial, CRD
 - Hierarchy is natural in some problems
 - Sampling - more concentrated than CRD

Question

- Example:
 - 10 schools in a school district (fixed effect)
 - Math scores from 20 randomly selected kids/school
- Draw a diagram for this example
- Do we see where the degrees of freedom error come from?
- If one factor is nested within another, can they interact?

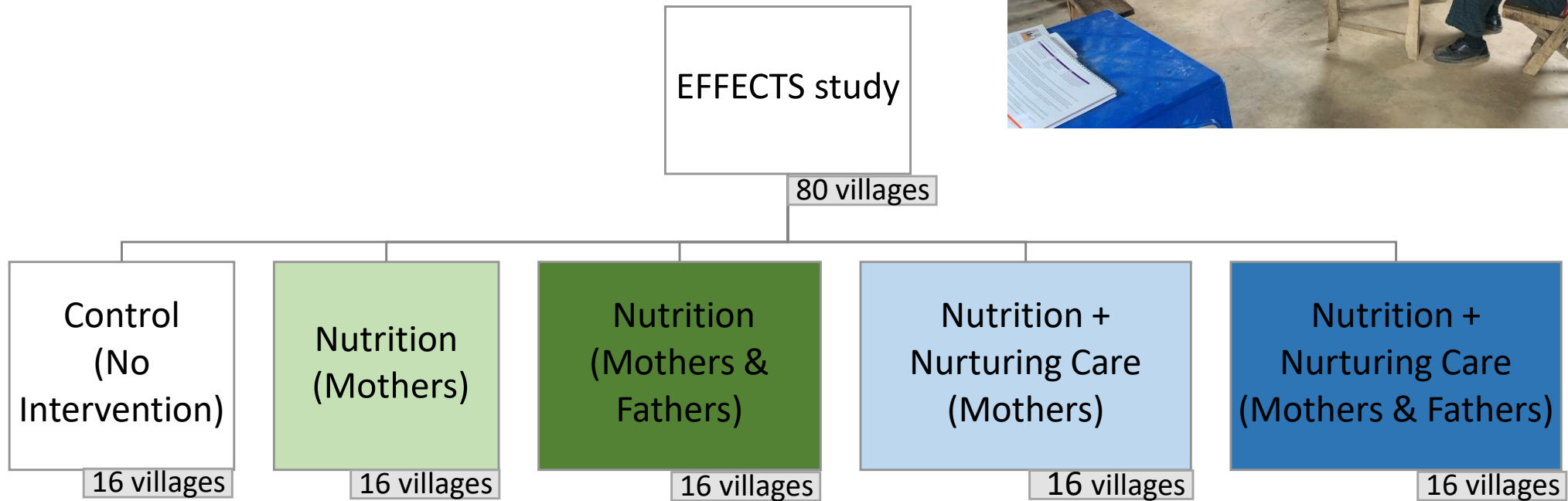
Cluster Randomized Trials

- Sometimes we may want to randomize groups of individuals to interventions
 - Group randomization vs. individual randomization
- Why would we want to do this?
 - Nature of the interventions under study
 - Difficulties in implementation
 - Participants within a group won't accept different treatments
 - Spillover between treatment groups
 - Ethical issues may or may not be addressed
- The level of randomization and the level of measurement are often not the same

Cluster RCT Examples

- Classrooms randomized to interventions, but children within those classrooms are measured
- Community-based interventions are randomized at community level, but assessments are made on individuals within those communities

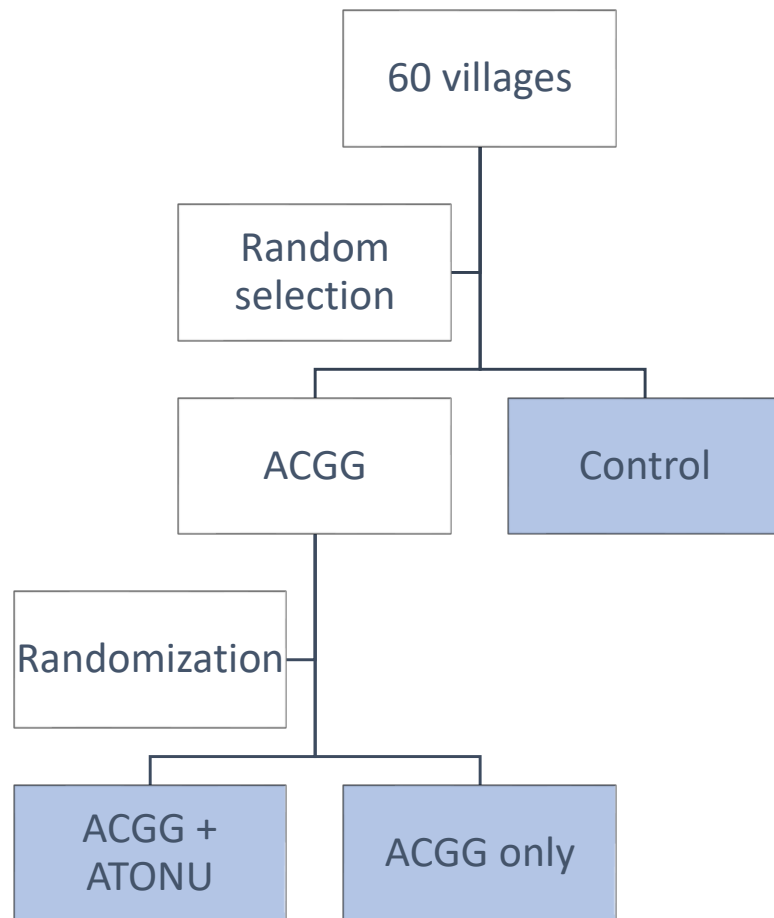
Example: EFFECTS Study



- Concern: spillover

ATONU Study - Design

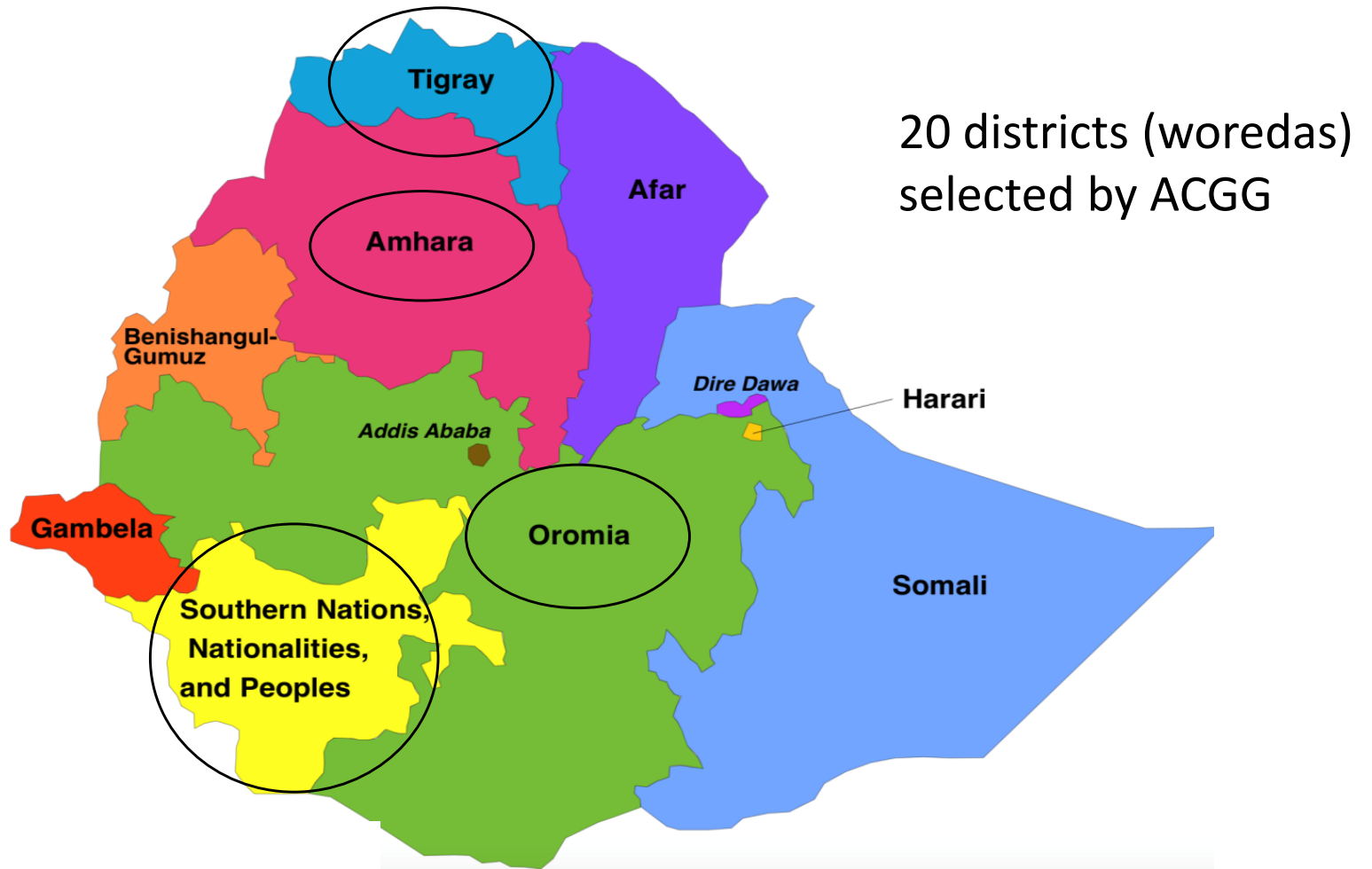
- Cluster randomized design in which villages are allocated to one of three treatment groups:



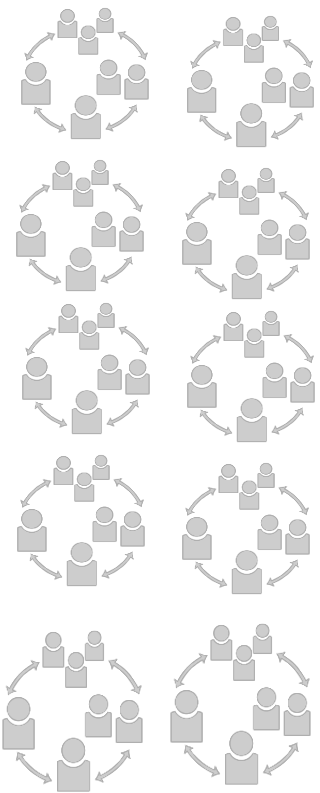
Comparisons of interest :

- ACGG only vs. Control
- ACGG + ATONU vs. Control
- ACGG +ATONU vs.ACGG only

Regional Map of Ethiopia



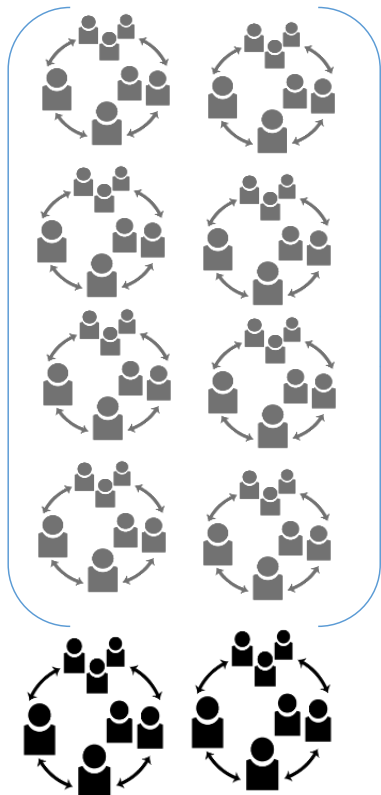
Village Sampling and Randomization



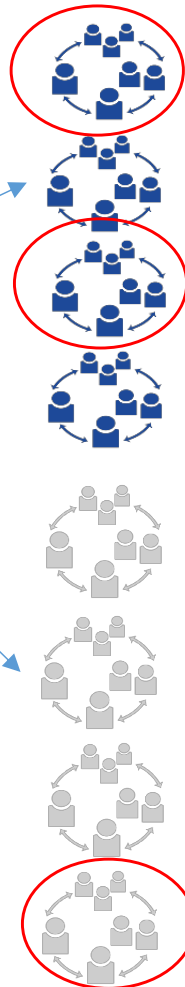
 = village

Village Sampling and Randomization

Agro-ecological zones



Selected for ACGG?



Villages (kebeles)



ACGG



ACGG + ATONU

Once a village is selected, households within a village can be randomly selected, screened for whether they meet inclusion/exclusion criteria, and enrolled



Control

 = village

Statistical Model

$$y_{ijk} = \mu + \tau_i + \beta_{j(i)} + \epsilon_{k(ij)}$$

y_{ijk} = observed value for the k th observation within the j th group of the i th treatment

μ = grand mean

τ_i = i th treatment effect

$\beta_{j(i)}$ = j th (random) group effect, nested in treatment i

$\epsilon_{k(ij)} \sim \text{iid } N(0, \sigma^2)$

$\beta_{j(i)} \sim \text{iid } N(0, \sigma_\beta^2)$

$\epsilon_{k(ij)}$ and $\beta_{j(i)}$ are independent

“hfias cluster randomization.sas”

- 20 villages, randomized:
 - 10 receiving food security intervention
 - 10 receiving no intervention
- 12 randomly selected households/village
- In each household, measure food insecurity (HFIAS score)
 - 0=food secure
 - 27=very food insecure

Cluster RCTs: Statistical Considerations

- Your research question is at cluster level
- Your statistical power comes from the number of clusters
 - Whenever possible, have more clusters, rather than people within a cluster
 - Easier said than done
- The more similar individuals are within a cluster, the less informative those individuals are
 - The lack of independence among individuals in the same cluster: [intracluster correlation](#)

Quantifying the Effect of Clustering

- Consider a trial w/ an experimental and control group, denoted by $i=1$ and 2 , respectively
- Each group has k clusters of size m
- $\bar{Y}_i$ is the mean response in group i
- Then:

$$\text{var}(\bar{Y}_i) = \frac{\sigma^2}{km} [1 + (m-1)\rho], \quad i = 1, 2$$

Where ρ is the intraclass correlation coefficient (ICC)

- $-1 \leq \rho \leq 1$, usually positive

Consequences of Cluster Randomization

- Sample size calculations for an individually randomized study will lead to an underpowered cluster randomized study → type II error
- Analysis ignoring clustering will underestimate standard errors → smaller p-values → type I error

Cluster RCTs: Statistical Considerations

- If you have very few clusters, your study may be considered quasi-experimental rather than experimental/randomized
- If you have 2 interventions, 2 villages each, what's the chance that all villages in one intervention will have higher average outcomes than all villages in the other intervention?

Rankings Just By Chance.... (24 possible rankings)

Treatment 1		Treatment 2	
Village 1	Village 2	Village 3	Village 4
1	2	3	4
1	2	4	3
1	3	2	4
1	3	4	2
1	4	2	3
1	4	3	2
2	1	3	4
2	1	4	3
2	3	1	4
2	3	4	1
2	4	1	3
2	4	3	1
3	1	2	4
3	1	4	2
3	2	1	4
3	2	4	1
3	4	1	2
3	4	2	1
4	1	2	3
4	1	3	2
4	2	1	3
4	2	3	1
4	3	1	2
4	3	2	1

Treatment 1		Treatment 2		
Village 1	Village 2	Village 3	Village 4	
1	2	3	4	*
1	2	4	3	*
1	3	2	4	
1	3	4	2	
1	4	2	3	
1	4	3	2	
2	1	3	4	*
2	1	4	3	*
2	3	1	4	
2	3	4	1	
2	4	1	3	
2	4	3	1	
3	1	2	4	
3	1	4	2	
3	2	1	4	
3	2	4	1	
3	4	1	2	*
3	4	2	1	*
4	1	2	3	
4	1	3	2	
4	2	1	3	
4	2	3	1	
4	3	1	2	*
4	3	2	1	*

8/24 possible rankings (33%!) have both villages in one treatment ranking higher than both villages in the other treatment

Selecting the Unit of Inference

- Do you want to make inferences about individuals or clusters?
 - This affects both design and analysis
- Examples (courtesy of A. Donner)
 - An intervention consists of a blood pressure screening program intended to lower the risk of cardiovascular mortality
 - A trial randomizing hospitals aims to assess the effect of obtaining a second clinical opinion on the decision to proceed with a caesarian section
 - The target of the intervention was the hospital rate of caesarian section
 - The hospital was the natural unit of inference
 - Apply standard methods of sample size determination and data analysis at the cluster level

More Design Considerations

- In many cluster RCTs, blinding is not possible (the nature of the intervention that necessitates cluster randomization may also make blinding hard)
- If recruitment occurs after randomization, could treatment assignment affect recruitment? (potential selection bias)
 - Example: physicians are randomized, then asked to recruit eligible patients
 - Possible solutions:
 - Ask physicians to ID eligible participants before randomization
 - Have a neutral party recruit participants

CONSORT Guidelines for Cluster RCTs

- <https://www.consort-spirit.org/extensions>
 - <https://www.bmj.com/content/bmj/345/bmj.e5661.full.pdf>
- Includes:
 - Rationale for adopting a cluster design
 - Incorporation of clustering into sample size calculation and analysis
 - Chart showing flow of clusters through the trial, from randomization to data analysis

Variation: Stepped-wedge Design

- All clusters eventually cross over from the control to the intervention at a time point determined at random
- Allows clusters to be enrolled gradually over time
- Allows RCT while addressing some potential ethical concerns

Variation: Stepped-wedge Design

- Betrán et al. 2017: Provision of medical supply kits to improve quality of antenatal care in Mozambique: a stepped-wedge cluster randomised trial

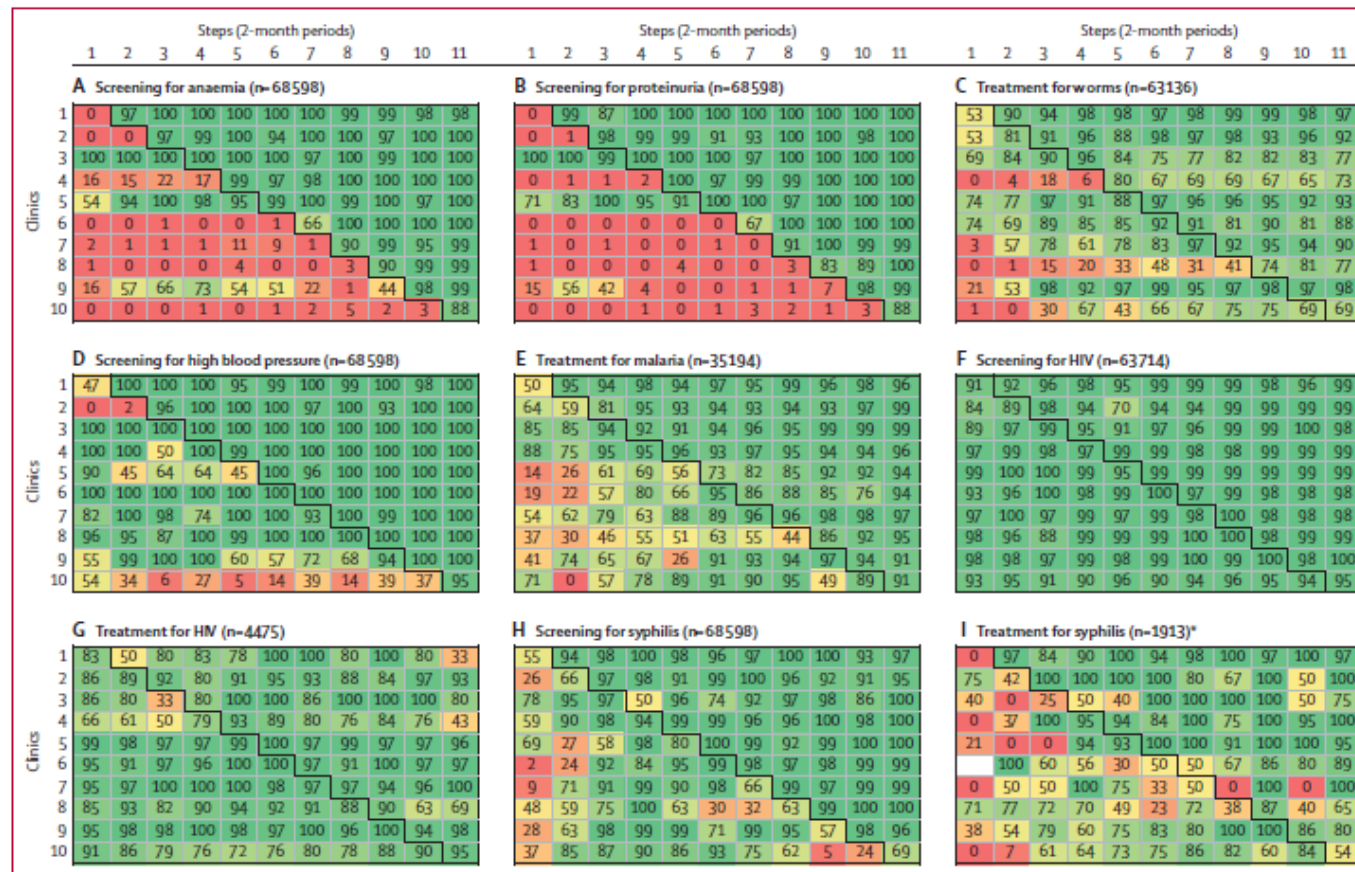


Figure 2: Outcome rates, by step and health facility; first antenatal care visits

Each cell contains the proportion of women who received each specific antenatal care practice in the corresponding antenatal care clinics and steps. For each of the nine panels in this figure, the ten clinics are represented in the y-axis while the steps of the trial are represented in the x-axis. *When the denominator to compute the cell rate is 0, cells are coloured in white.